

HDOC - Day 17

Digit Rearrangement, Swapping and Palindrome Using Void Functions

Algorithm

1. Start. 2. Display menu. 3. Call a separate parameterless void function for the selected option. 4. Largest number: count digit frequencies and append digits from 9 down to 0. 5. Smallest number: place the smallest non-zero digit first, then zeros, then remaining digits in ascending order. 6. Swap first/last: determine the highest place value and reconstruct the number with exchanged end digits. 7. Palindrome: reverse the number and compare it with the original. 8. Repeat until Exit. 9. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	31042; option 1	Largest=43210	43210	Pass
2	31042; option 2	Smallest=10234	10234	Pass
3	12345; option 3	52341	52341	Pass
4	12321; option 4	Palindrome	Palindrome	Pass

C Program

```
#include <stdio.h>
void largestNumber(void); void smallestNumber(void); void swapFirstLast(void); void palindrome(void);
int main() {
    int ch;
    do {
        printf("\n1. Largest possible number\n2. Smallest possible number\n");
        printf("3. Swap first and last digits\n4. Palindrome check\n5. Exit\nEnter choice: ");
        scanf("%d",&ch);
        switch(ch) {
            case 1: largestNumber(); break; case 2: smallestNumber(); break;
            case 3: swapFirstLast(); break; case 4: palindrome(); break;
            case 5: break; default: printf("Invalid choice.\n");
        }
    } while(ch!=5); return 0;
}

void largestNumber(void) {
    int n,f[10]={0},d,i; long result=0;
    printf("Enter a positive integer: "); scanf("%d",&n);
    while(n>0){f[n%10]++;n/=10;}
    for(i=9;i>=0;i--) while(f[i]-- > 0) result=result*10+i;
    printf("Largest possible number = %ld\n",result);
}

void smallestNumber(void) {
    int n,f[10]={0},i; long result=0;
    printf("Enter a positive integer: "); scanf("%d",&n);
    while(n>0){f[n%10]++;n/=10;}
    for(i=1;i<=9;i++) if(f[i]) { result=i; f[i]--; break; }
    while(f[0]-- > 0) result*=10;
    for(i=1;i<=9;i++) while(f[i]-- > 0) result=result*10+i;
    printf("Smallest possible number = %ld\n",result);
}

void swapFirstLast(void) {
    int n,temp,last,first,p=1,middle; printf("Enter a positive integer: "); scanf("%d",&n);
    if(n<10){printf("After swapping = %d\n",n);return;}
    last=n%10; temp=n;
    while(temp>=10){temp/=10;p*=10;} first=temp;
    middle=(n%p)/10;
    printf("After swapping = %d\n",last*p+middle*10+first);
}

void palindrome(void) {
    int n,temp,rev=0,d; printf("Enter a positive integer: "); scanf("%d",&n); temp=n;
    while(temp>0){d=temp%10;rev=rev*10+d;temp/=10;}
    printf(n==rev?"Palindrome number\n":"Not a palindrome number\n");
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day17.c -o day17
puppie@Anvikas-MacBook-Air Desktop % ./day17
Enter choice: 1
Enter a positive integer: 31042
Largest possible number = 43210
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day17
Enter choice: 2
Enter a positive integer: 31042
Smallest possible number = 10234
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day17
Enter choice: 3
Enter a positive integer: 12345
After swapping = 52341
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day17
Enter choice: 4
Enter a positive integer: 12321
Palindrome number
```

Result: The menu-driven C program was compiled, executed and tested successfully against the prepared test cases.